

1: Introduction

In a wind tunnel, air of velocity v passes a certain model, and the air resistance force is measured directly. In the present notes it is shown how it can also be found indirectly by measuring the acceleration the model gets when it passes through the air, mounted on an arm, protruding from a "merry-go-round" (a bicycle wheel) that rotates around a vertically axis. Velocities measured by a photo gate are fed into a computer which compares the results of two runs, one with the model and one without.

For a model about 10 cm in size, moving with a velocity v in the range from 1,5 m/s to 10 m/s , the acceleration in each run is found to of the form $a = a_1v + a_2v^2$ where the two coefficients a_1 and a_2 are determined by extraordinarily good least squares fits. The air resistance force F on the model resulting from the two fits is not quite as accurate, because the inertial mass slowed down by the air resistance is not determined quite as accurately. Furthermore, F is determined as the difference between two terms determined separately. It is, however, still rather good. In such well described cases as a smooth sphere and a circular disk, one gets the expected drag values within 2-3%. Students, however, are more likely to be interested in the following experiments.

In the shop with gifts for "the man who has everything", one can buy beautiful model cars, rather meticulously built in 1:18. Determined by the merry-go-round, the drag coefficient for some of these - from vintage car to modern sport car - turns out to fit reasonably well with the full scale cars. It is not *a priori* obvious that C_d for the small models, going at only 10 m/s, should fit as well as they do. In secs. 7 and 8 we discuss whether it is more than just a coincidence.

Fitting a model rocket on the merry-go-round, one can measure not only its drag coefficient C_d , but also its velocity and acceleration as functions of time. Thereby one can determine the reaction force from its motor as a function of time, measured from the moment of ignition. These data allows a check of the specifications for the motor and together with the rockets C_d , one has the data for a computer simulation of a real launch.

The merry-go-round presents the student to a variety challenges - theoretical as well as experimental. Unexperienced students can understand the basic idea and make the experiments if only they are willing to consider certain formulas as "facts". More advanced students can set up and solve the differential equations for the motion of the models. They can learn about the physical meaning of the moment of inertia for the merry-go-round. They can question the meaning of the extremely accurate fitting results - do these actually lead to a correspondingly accurate determination of the parameters we are going for?. Also the students who wants to dig deeper, can get a motivating experimentally based introduction to Reynolds fascinating theory for scaling between models of various sizes, moving with different velocities through various media.

We proceed as follows. Section 2 gives a first introduction to the meaning of the drag coefficient C_d . In sec. 3 the merry-go-round is described and examples are given on how unusually well - almost *incredibly* well - the decreasing velocity v of a model can be fitted to an acceleration which can be written as

$$a = -\alpha v - \beta v^2 \quad (1.1)$$

where α and β are constants. Then, in sec. 4, it is shown how the accelerations for the two runs considered in sec. 3 can be used to determine how the force of the air resistance on the model depends on the velocity. It follows directly how the drag coefficient, within the interval of velocities considered, takes the form

$$C_d = \frac{c_1}{v} + c_2 \quad (1.2)$$

where c_1 is a constant velocity and c_2 is a dimensionless constant. Often it is this latter constant one refers to when talking about the drag coefficient - a usage justified in cases where term $\frac{c_1}{v}$ is small. The actual values of the two constants in (1.2) do not follow directly from the values of α and β in the acceleration (1.1): According to Newton's 2. law, a mass is needed to come from acceleration to force. *We must determine the inertial mass that is being slowed down by the air resistance force.* Sec. 5 explains how to deal with this problem and then, in the subsequent sections, we look at the determination of C_d from some concrete measurements. In sec. 6 we look at the results for a sphere and a circular disk, and see that the results come out in fine agreement with the well established results found in the literature. In sec. 7 we look at C_d for some car models and discuss the rather good agreement with the values for the full scale models.

In sec. 8 we turn to Reynolds theory and learn why we cannot in general expect our C_d -values for the car models to agree with those of full scale cars, going much faster (although, in the cases considered, they actually do so quite well).

In sec. 9 we briefly explain how the merry-go-round also has been used to measure the thrust of a model rocket as a function of the time.

In the appendices A and B we look closer into the mathematical theory behind the merry-go-round. Thus the expression (1.1) for the acceleration leads to the equation of motion

$$\frac{dv}{dt} = -\alpha v - \beta v^2 \quad (1.3)$$

and in appendix A we solve and discuss this differential equation from various points of view. Sec. 5 explains in intuitive terms how the equation of motion for a model on the merry-go-round is the same as that for a hypothetical "equivalent car" which moves along a linear s-axis instead of the circular one. The strict mathematical explanation, involving the moment of inertia of the merry-go-round, is given in appendix B. In appendix C a spreadsheet is presented by which one can check the accuracy of the fits described in the main text. The spreadsheet

also calculates good values from which to start the fitting process. Such values can be determined by guessing many times, but this is rather tedious. The formula for the good start values is derived in appendix D.

The graphs and the tables shown in the main text are found by use of a program Cd.fpr, written in FPro, a simple but most useful program for mathematics, developed by Danish teachers. Since this is not known internationally, the notes do not contain a documentation of Cd.fpr. However, with one exception, all the essentials can also be made by a spreadsheet like the one in appendix C. The exception is, that one must have a programme that can determine the least squares fit described in the eqns. (3.1)-(3.3). Least squares fitting is a standard procedure in FPro - it does not yet seem to be so in familiar pocket calculators like TI89 even though these are much stronger in many other respects. That is regrettable.